One Seed for Every Adaptive Policy: Exact Causal Simulation and Sharp Entropy Asymptotics

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Abstract

We study the least entropy of a random seed that simulates a controlled history tree exactly under every deterministic adaptive policy, using a single decoder consistent at shared histories. For a finite tree, let L be the mean of the pointwise lower envelope of the policy information CDFs. A causal greedy construction gives $L \le C \le L + c_\star$, where $c_\star = (1+\log_2 e)/2$. We extend the result to locally finite infinite trees: $C_\infty = \sup_n C_n$, with attainment whenever this value is finite. We also identify the attained atomic capacity and construct one countable causal seed satisfying simultaneous Rényi bounds. For stationary finite kernels with common entropy h, the excess is $\sqrt{2/\pi}(\sigma_{\max} - \sigma_{\min})\sqrt{n} + o(\sqrt{n})$. When the information variances coincide at $v > 0$, we prove the finer law

$$C_n = nh + \frac{\kappa_{\max} - \kappa_{\min}}{3v\sqrt{2\pi e}} \ln n + O(1),$$

where the κ_i are centered third information moments. The proof combines an explicit Gaussian corrector with uniformly bounded integrated Bellman amplification. No lattice assumption is required. The main statements are linked to their Lean formalizations in a fixed repository revision. The numerical additive constant and the nonlinear diffusion profile are attributed to prior work.

Keywords: exact simulation; shared randomness; adaptive policies; minimum entropy coupling; information spectrum; Bellman equation; formal verification.

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1 Introduction and main results

A simulator receives an action, produces a random response, and repeats. The next action may depend on the entire observed history. We ask how much randomness must be drawn at the start so that every later response is a deterministic function of that seed, the history, and the current action. Exactness is required for every policy, with the same seed law and decoder.

The compatibility requirement is substantive. Coupling the transcript laws of all policies does not, by itself, ensure that two policies receive the same response when they have the same history and choose the same action. We impose this agreement atom by atom by taking seed values to specify total response strategies. Entropy measures the information in the initial seed, not the size of a representation of the decoder or its running time.

1.1 The finite theorem

For a finite controlled tree T, let I_π be the information, in bits, of its random transcript under policy π. Write

$$F_*(t) = \min_\pi \mathbb{P}(I_\pi \le t), \qquad L(T) = \int_0^\infty (1 - F_*(t))\, dt.$$

Let $C(T)$ be the minimum entropy of an exact causal seed. A seed may be finite or countable; these give the same optimum on a finite tree.

Theorem 1.1 (Finite causal Shannon theorem)**.** *Every finite controlled tree admits a finite exact causal seed S_G with*

$$L(T) \le C(T) \le H(S_G) \le L(T) + c_\star, \qquad c_\star = \frac{1 + \log_2 e}{2}. \tag{1.1}$$

More precisely, if Z has CDF F_* and J_G is the information of S_G, then

$$Z \preceq_{\mathrm{st}} J_G \preceq_{\mathrm{st}} Z + W, \qquad W = A(1+E), \quad A \sim \mathrm{Bernoulli}(1/2), \quad E \sim \mathrm{Exp}(\ln 2), \tag{1.2}$$

where Z, A, E on the right are independent.

The bound is independent of depth and of all finite local alphabet sizes. The constant $c_\star$ is the ordinary greedy-coupling constant of Compton et al. [1]; the proof below supplies a compatible strategy construction on a controlled tree.

1.2 Infinite horizons

Let T_n be the depth-n truncation of a locally finite tree, and let C_∞ be the infimum of the entropy of one countable seed exact at every finite horizon. An empty feasible class has cost $+\infty$.

Theorem 1.2 (Projective identity)**.**

$$C_\infty = \sup_n C(T_n), \qquad L_\infty \le C_\infty \le L_\infty + c_\star, \quad L_\infty = \sup_n L(T_n).$$

If $C_\infty < \infty$, a countable exact seed attains the infimum. Consequently a finite-entropy exact seed exists if and only if $\sup_n L(T_n) < \infty$.

Section 3 also treats atomic capacity when the best exact representation has a nonatomic component. Section 4 gives one countable causal seed with simultaneous bounds for every positive Rényi order. These are distinct assertions: countable support need not imply finite Shannon entropy, and a simultaneous approximation need not attain the Shannon minimum.

1.3 The stationary hierarchy

Suppose a fixed finite family (P_i) is available at every history, with common entropy h. Put

$$X_i = -\log_2 P_i(Y_i) - h, \quad Y_i \sim P_i, \quad \sigma_i^2 = \mathbb{E}X_i^2, \quad \kappa_i = \mathbb{E}X_i^3.$$

Let C_n denote the optimal seed entropy for n rounds.

Theorem 1.3 (Stationary entropy asymptotics)**.** *Set $a = \max_i \sigma_i$ and $b = \min_i \sigma_i$.*

1. *For $a > b \geq 0$,*
$$C_n = nh + \sqrt{2/\pi}(a-b)\sqrt{n} + o(\sqrt{n}).$$
If $b > 0$, the remainder is $O(n^{1/4}\ln n)$.

2. *If every $\sigma_i^2 = v > 0$, then*
$$\boxed{C_n = nh + c\ln n + O(1), \qquad c = \frac{\kappa_+ - \kappa_-}{3v\sqrt{2\pi e}},} \tag{1.3}$$
where $\kappa_+ = \max_i \kappa_i$ and $\kappa_- = \min_i \kappa_i$.

3. *If $a = b = 0$, then $C_n = nh$ for every n.*

All asymptotic constants refer to a fixed kernel. In parts 1 and 2, the same expansion holds for the envelope mean and the greedy seed entropy.

The third assertion is elementary: zero information variance means each law is uniform on its support. Equal entropy makes all support sizes the same integer m. A sequence of n independent uniform labels in $\{1,\dots,m\}$, relabeled for each action, is exact with entropy $n\log_2 m$. The fixed-policy entropy lower bound gives equality.

For $v > 0$, the first three centered information moments therefore determine which of the following orders occurs:

Moment condition (common entropy)	Excess $C_n - nh$
Different variances	$\Theta(\sqrt{n})$
Common variance, different third moments	$c\ln n + O(1)$, $c > 0$
Common variance and third moment	$O(1)$

The last row is a boundedness assertion, not a positive lower bound or a claim that the bounded remainder converges.

1.4 Relation to previous work

Minimum entropy coupling and functional representation provide the ordinary, marginal version of the problem. Li [3] treats approximate couplings for multiple, including infinite, families of distributions. Shkel and Yadav [2] develop information-spectrum converses. Compton et al. [1] obtain the additive constant $c_\star$ through the profile method. We use that constant with explicit attribution and prove the requisite residual-mass estimate for compatible tree strategies. A later approximation scheme of Compton [4] concerns the ordinary finite-marginal optimization problem; our construction does not assert a polynomial running time in a succinct description of the controlled tree.

The controlled diffusion underlying the square-root regime belongs to the nonlinear central limit theory of Peng [5]. The one-sided capacity profile appears in Peng and Zhou [6]; convergence rates under sublinear expectations are studied by Fang et al. [7]. We include a

discrete comparison proof to establish the precise connection with causal seed entropy and to make the passage to means explicit. The equal-variance analysis instead resolves the logarithmic displacement through a third-moment corrector and a bounded amplification lemma. We do not claim historical novelty for the ordinary coupling bound, the nonlinear heat profile, or the underlying information-spectrum converse.

1.5 Conventions and formalization

All entropies and information variables in this paper are in bits; ln is the natural logarithm. In particular c in (1.3) is measured per unit of $\ln n$. We use $0 \log 0 = 0$. Unless stated otherwise, local kernels are positive; Section 5 explains the extension that keeps zero branches.

The exposition is tied to repository revision `5741ee193bd5`. Section 8 maps the principal statements to Lean declarations and specifies what the recorded axiom audit establishes. This paper is an exposition of the selected main results, not a claim that every auxiliary result in the repository is reproduced here.

2 Finite controlled trees and exact seeds

At a history v, a finite nonempty set of actions is available. Each action a has a finite nonempty output set $\mathcal{Y}(v, a)$ and a probability kernel $K_v(y \mid a)$. The child history is vay. Put

$$p(\varnothing) = 1, \qquad p(vay) = p(v)K_v(y \mid a).$$

Terminal histories may occur at different depths. A policy selects an action at every nonterminal history; a response strategy d selects an output at every history-action pair. Together they determine one terminal transcript, denoted $\mathrm{run}(\pi, d)$.

Write $\chi_d(v)$ for the indicator that d agrees with every output along v. For each action, $\sum_y \chi_d(vay) = \chi_d(v)$. A seed with label weights (w_j) and response strategies (d_j) is exact when

$$\sum_j w_j \chi_{d_j}(v) = p(v) \qquad \text{for every history } v. \tag{2.1}$$

This is equivalent to the prescribed transcript law under every policy. For one direction sum over the terminal continuations of a history under a policy following its actions; for the other apply (2.1) to terminal histories compatible with the policy. Choices at histories incompatible with a strategy still belong to its total decoder.

There are finitely many strategies. Coarsening any countable seed by its strategy preserves exactness and cannot increase entropy. The exact laws on strategies form a nonempty compact polytope. Nonemptiness follows by independently sampling a response at each history-action pair. Continuity of finite Shannon entropy shows that $C(T)$ is a minimum.

2.1 An information-spectrum converse

For every policy, a seed atom of mass w is mapped into a transcript atom of mass $p \geq w$. Thus seed information J dominates transcript information under the induced coupling. Consequently

$$F_J(t) \leq F_{I_\pi}(t) \quad \text{for every } \pi, \qquad F_J(t) \leq F_*(t). \tag{2.2}$$

The minimum F_* is a CDF because there are finitely many policies. Integration gives $H(S) \geq L(T)$. This converse also holds for ordinary couplings of the policy transcripts without shared-history consistency. For any exact seed with finite entropy, the one-dimensional transport identity gives

$$W_1(\mathcal{L}(J), \mathcal{L}(Z)) = \int_{\mathbb{R}} |F_J - F_*| = H(S) - L(T). \tag{2.3}$$

2.2 Bottleneck minimax and extraction

A residual flow r is nonnegative and satisfies $\sum_y r(vay) = r(v)$ for every action. Define

$$B_r(v) = r(v) \quad (v \text{ terminal}), \qquad B_r(v) = \min_a \max_y B_r(vay) \quad (v \text{ nonterminal}).$$

Lemma 2.1 (Tree minimax)**.** *If $\mathcal{L}(d)$ and $\mathcal{L}_\pi$ denote the leaves compatible with strategy d and policy π, respectively, then*

$$\max_d \min_{v \in \mathcal{L}(d)} r(v) = B_r(\varnothing) = \min_\pi \max_{v \in \mathcal{L}_\pi} r(v).$$

This is the largest w for which $r - w\chi_d$ is a nonnegative flow for some strategy d.

Proof. Induct on the tree. A strategy chooses an output for each action and a continuation on every selected child. Maximizing its smallest compatible leaf mass gives $\min_a \max_y B_r(vay)$, since the choices for distinct actions are independent. A policy chooses one action and continuations for every output, giving the same recursion for its smallest achievable largest leaf mass. The leaf case is immediate. Nonnegativity on the leaves implies nonnegativity on every cylinder by flow conservation. Subtracting $w\chi_d$ preserves that conservation. This proves the last assertion as well. □

Starting with $r = p$, extract a maximizing strategy with weight $w = B_r(\varnothing)$ and subtract $w\chi_d$. Positive root mass gives positive bottleneck by induction. At least one previously positive leaf becomes zero. Hence extraction terminates in at most the number of leaves of the full controlled tree. The extracted weights are nonincreasing, sum to one, and satisfy (2.1). This is the seed S_G.

2.3 The refined cutoff and the universal constant

Fix $0 < \delta \le 1$ and stop just after all weights greater than δ have been extracted. The remaining root mass is $R = \sum_{j:w_j \le \delta} w_j$. By Lemma 2.1, some policy has every residual leaf mass at most δ. On this policy a leaf with original mass $p > \delta$ must already have lost a weight greater than δ. Its residual is therefore at most

$$a_\delta(p) = \begin{cases} p, & p \le \delta, \\ \min\{\delta, p - \delta\}, & p > \delta. \end{cases}$$

Define the nondecreasing function

$$\psi(u) = \begin{cases} u, & 0 \le u \le 1/2, \\ 1/2, & 1/2 < u < 1, \\ 1, & u \ge 1. \end{cases}$$

For every $p > 0$, $a_\delta(p) \le p\psi(\delta/p)$: equality holds outside $\delta < p < 2\delta$, and on that interval the inequality is $p - \delta \le p/2$. Thus

$$\begin{aligned} \sum_{j:w_j \le \delta} w_j &\le \max_\pi \sum_{v \in \mathcal{L}_\pi} a_\delta(p(v)) \\ &\le \max_\pi \mathbb{E}\psi(\delta 2^{I_\pi}) \le \mathbb{E}\psi(\delta 2^Z). \end{aligned} \tag{2.4}$$

The last step uses $I_\pi \preceq_{\mathrm{st}} Z$. It is distinct from the preceding pointwise majorization: $a_\delta(p)/p$ itself is not monotone in information.

For the variable W in (1.2), its inclusive survival at s is 1 for $s \le 0$, $1/2$ for $0 < s < 1$, and 2^{-s} for $s \ge 1$. Setting $\delta = 2^{-t}$ in (2.4) therefore gives

$$\mathbb{P}(J_G \ge t) \le \mathbb{E}\psi(2^{Z-t}) = \mathbb{P}(Z + W \ge t).$$

Passing to thresholds decreasing to t proves the strict-tail inequality, including atoms. Together with (2.2), this proves (1.2). Finally $\mathbb{E}W = (1 + 1/\ln 2)/2 = c_\star$, completing Theorem 1.1.

Remark 2.2. The constant is a universal upper bound, not an overhead paid by every instance. If all actions request one fixed law, up to bijective relabeling, an independent sequence with that law is an optimal seed and $C_n = nh$. Nor does the theorem assert that a minimum-entropy ordinary coupling is compatible; it constructs a separate causal seed.

3 Infinite horizons and atomic capacity

3.1 Strategy space and capacity

Let T be a rooted controlled history tree. At every finite history h there are finitely many available actions, each with a finite nonempty output alphabet and a prescribed probability kernel. We assume the alphabets are finite at every node, not uniformly bounded. Terminal nodes can be padded by a forced action and output. Zero-probability outputs may be deleted; deterministic singleton alphabets are permitted.

Write $p(h)$ for the product of output probabilities along h. For every available action a,

$$p(\varnothing) = 1, \qquad \sum_y p(hay) = p(h).$$

At each finite depth there are finitely many histories. A deterministic response strategy selects one output at every history–action pair. The space $\mathcal{S}$ of all such strategies is a countable product of finite discrete spaces, hence compact and metrizable. For a history h, let $E_h \subseteq \mathcal{S}$ be the set of strategies agreeing with its output choices. This is a clopen cylinder, and

$$E_h = \bigsqcup_y E_{hay} \quad \text{for every action } a.$$

A Borel probability measure μ on $\mathcal{S}$ is an exact causal representation if $\mu(E_h) = p(h)$ for every finite history h. This is equivalent to exactness for every deterministic adaptive policy at every finite horizon, and therefore for the corresponding infinite path laws.

Let $P_{n,\pi}$ be the finite transcript law at horizon n under policy π. Put

$$A_K(P) = \max_{D:\,|D|\le K} P(D), \tag{3.1}$$

$$a_K(T) = \inf_{n,\pi} A_K(P_{n,\pi}), \qquad q(T) = \lim_{K\to\infty} a_K(T). \tag{3.2}$$

The limit exists by monotonicity and belongs to $[0, 1]$. For a Borel probability on a compact metrizable space define its total atomic mass by

$$\mathrm{At}(\mu) = \sum_{s:\,\mu(\{s\})>0} \mu(\{s\}).$$

There are at most countably many terms.

Ordinary couplings below are Borel probability laws on the countable product of the transcript spaces indexed by all pairs (n, π), with the specified coordinate marginals. They need not be compatible with a single strategy or even consistent between horizons.

Theorem 3.1 (Atomic capacity)**.** *For every such tree,*

$$\max_{\mu \text{ exact causal}} \mathrm{At}(\mu) = \max_{\Gamma \text{ ordinary}} \mathrm{At}(\Gamma) = q(T).$$

Both maxima are attained. In particular, an exact countably supported causal seed for the entire infinite tree exists if and only if $q(T) = 1$. If $q(T) < 1$, every exact causal representation has nonatomic mass at least $1 - q(T)$, and a representation attaining this minimum exists.

3.2 A finite concentration estimate

We use the bottleneck construction proved in Section 2. After a finite number of causal greedy extractions, the residual cylinder masses form a nonnegative controlled flow. If its root mass is R, its next extractable strategy weight w equals the smallest, over policies, of the largest residual transcript mass. Extraction terminates on each finite tree and gives an exact finite seed.

Lemma 3.2. *Suppose every policy on a finite controlled tree has K transcripts carrying total mass at least a. The first M greedy atoms, padded with zeros if extraction has ended, have total mass at least*

$$a\big[1 - (1 - 1/K)^M\big].$$

Consequently the same is true of the M largest seed atoms.

Proof. Let R_m be the residual root mass after m steps, with $R_0 = 1$. The minimax identity supplies a policy all of whose residual transcript masses are at most the next weight w_{m+1}. On this policy select at most K original transcripts of mass at least a. The residual mass outside this set is at most $1 - a$, so

$$R_m - (1 - a) \le K w_{m+1}.$$

Since $R_{m+1} = R_m - w_{m+1}$,

$$R_{m+1} - (1 - a) \le (1 - 1/K)(R_m - (1 - a)).$$

Iteration gives $R_M \le 1 - a + a(1 - 1/K)^M$. The same bound holds after termination. Subtract from one. For $K = 1$ and $M \ge 1$ the displayed formula has its usual value a. □

3.3 Attainment of atomic capacity

Proof of Theorem 3.1. Upper bound, even without causality. Take any m atoms of an ordinary coupling, of total mass b. Their images in any one transcript coordinate lie in at most m transcripts, so $b \le A_m(P_{n,\pi})$ for every (n, π). Hence $b \le a_m \le q$. Exhausting all its atoms gives $\mathrm{At}(\Gamma) \le q$. The same argument applies directly to a causal representation by projecting each atom through a policy.

Limits of weights and strategies. For each horizon n, take a finite greedy causal seed, order its weights as $w_1^{(n)} \ge w_2^{(n)} \ge \cdots \ge 0$, and pad by zeros. Extend each finite strategy arbitrarily to an element $s_i^{(n)}$ of $\mathcal{S}$. By compactness and diagonal selection there is one subsequence $n_j \to \infty$ such that, for every i,

$$w_i^{(n_j)} \to w_i, \qquad s_i^{(n_j)} \to s_i.$$

Set $\nu = \sum_i w_i \delta_{s_i}$ and $A = \sum_i w_i \le 1$; repeated limit strategies are simply merged as measures. For fixed K, M, Lemma 3.2, with $a = a_K(T)$, gives

$$\sum_{i=1}^{M} w_i \ge a_K(T)\big[1 - (1 - 1/K)^M\big].$$

First let $M \to \infty$ and then $K \to \infty$. It follows that $A \geq q$.

No invalid passage of infinitely many atoms through a limit. For fixed h and M, convergence of strategies makes membership in the clopen set E_h eventually constant for each of the first M indices. For large j, the finite seed is exact at h. Thus

$$\sum_{i=1}^{M} w_i \mathbf{1}_{E_h}(s_i) = \lim_j \sum_{i=1}^{M} w_i^{(n_j)} \mathbf{1}_{E_h}(s_i^{(n_j)}) \leq p(h).$$

Monotone convergence in M proves $\nu(E_h) \leq p(h)$. Define

$$r(h) = p(h) - \nu(E_h) \geq 0.$$

Since the E_{hay} partition E_h for each action, $\sum_y r(hay) = r(h)$ and $r(\varnothing) = 1 - A$.

Realization of the missing mass. If $A < 1$, assign, independently at every history–action pair, an output with probabilities $r(hay)/r(h)$ whenever $r(h) > 0$, and any distribution when $r(h) = 0$. The countable product law λ of these response choices satisfies

$$\lambda(E_h) = r(h)/(1 - A).$$

Indeed this follows by induction along h from the displayed ratios; after a zero residual prefix both sides vanish. Hence $\mu = \nu + (1 - A)\lambda$ is exact. If $A = 1$, simply set $\mu = \nu$. As ν is atomic of mass A, $\mathrm{At}(\mu) \geq A \geq q$. The upper bound already proved gives $\mathrm{At}(\mu) = A = q$. In particular, the residual law in this construction has no atoms when $q < 1$.

Finally map $s \in \mathcal{S}$ to its full tuple of finite-policy transcripts. This is a measurable map into the countable product of finite spaces. The image of μ is an ordinary coupling and has atomic mass at least q, since images of atoms remain atoms. The upper bound makes its atomic mass exactly q. This proves attainment on both sides. □

3.4 Information spectrum and entropy attainment

All logarithms in this section have base two. Define

$$F_n(t) = \min_{\pi} P_{n,\pi}\{-\log P_{n,\pi}(X) \leq t\}, \quad F_\infty(t) = \inf_n F_n(t), \quad t \geq 0.$$

Refining a transcript cannot reduce its information, and policies extend to longer horizons. Therefore $F_n(t)$ decreases with n. For an individual finite law P, writing its information CDF as F_P,

$$F_P(t) \leq A_{\lfloor 2^t \rfloor}(P), \qquad A_K(P) \leq F_P(t) + K2^{-t}.$$

Taking infima over all finite-policy laws proves

$$\boxed{q(T) = \lim_{t \to \infty} F_\infty(t).}$$

Thus atomic capacity is precisely the mass that does not escape to infinite information in the uniform policy envelope. This statement does not require treating a possibly defective F_∞ as a probability CDF.

Let C_n be the minimum entropy of an exact seed for the depth-n tree. Let C_∞ be the infimum entropy among countably supported exact seeds for the full tree, with value $+\infty$ if there is no such seed. Set

$$L_n = \int_0^\infty (1 - F_n(t))\, dt, \quad L_\infty = \sup_n L_n, \quad c_* = (1 + \log e)/2.$$

Theorem 3.3 (No extra entropy at the projective limit)**.** *In the extended real numbers,*

$$\boxed{C_\infty = \sup_n C_n.}$$

If the right-hand side is finite, the infimum on the left is attained. Consequently the finite Shannon theorem implies

$$L_\infty \le C_\infty \le L_\infty + c_*.$$

In particular, $L_\infty < \infty$ is equivalent to the existence of one finite-entropy exact causal seed for the entire infinite tree.

Proof. Restriction of a seed proves $C_n \le C_{n+1} \le C_\infty$. If $B = \sup_n C_n = +\infty$, this already proves the equality. Assume $B < \infty$ and choose a minimizing finite seed at each horizon. Such minimizers exist: one can minimize entropy over the compact finite-dimensional polytope of laws on finite strategies, merging identical strategies if necessary. Order and pad weights as above. For $i > K$, $w_i^{(n)} \le 1/(K+1)$, whence

$$\sum_{i>K} w_i^{(n)} \le \frac{B}{\log(K+1)}. \tag{*}$$

Diagonal compactness gives limiting weights and strategies. The uniform tail bound (*) forces $\sum_i w_i = 1$. The cylinder argument in the previous proof gives $\nu(E_h) \le p(h)$. Since both are probability flows, equality follows at each depth by summing over the leaves of any policy following h, or by passing the uniformly small tails directly through the limit.

The countable label seed i with masses w_i realizes these strategies. With $0 \log 0 = 0$, continuity of each summand and Fatou's lemma give

$$H(w) = \sum_i -w_i \log w_i \le \liminf_j H(w^{(n_j)}) = B.$$

Together with restriction this yields $C_\infty = B$, with attainment. Finally the finite theorem gives $L_n \le C_n \le L_n + c_*$; take suprema. No continuity of Shannon entropy on an unrestricted countable simplex is being assumed. □

Three regimes. If $q < 1$, every exact representation has an unavoidable nonatomic component. If $q = 1$ but $L_\infty = \infty$, a countable exact seed exists but all such seeds have infinite entropy. If $L_\infty < \infty$, an optimal finite-entropy countable seed exists. Countable support is not the same as finite support.

3.5 Pointwise versus uniform simulability

Consider a deterministic waiting chain. At waiting stage $m \ge 1$ the controller can either wait for the next stage or trigger an experiment. Triggering at stage m produces m independent fair bits and then stops; there are no choices during the experiment. A policy that never triggers has a deterministic infinite transcript. The tree is locally finite, with at most two actions and at most two outputs at every node.

Every individual deterministic policy therefore has a finite-support full-path law: either one path, or 2^m paths for some finite m. Nevertheless, for each fixed K, choosing arbitrarily large m yields

$$a_K \le K 2^{-m}, \qquad a_K = 0, \qquad q = 0.$$

Hence *every exact common representation is atomless*, even without a causality requirement. An elementary direct proof is that an atom of mass $w > 0$ would have to map into a transcript

of mass $2^{-m} < w$ for a sufficiently large triggered experiment. This example illustrates why separate finite-support simulations do not imply one discrete seed. It is not claimed as a new counterexample in the general theory of ordinary coupling.

For any prescribed $q_0 \in (0,1)$, first choose a deterministic absorbing branch with probability q_0 and the waiting construction with probability $1 - q_0$. Then $a_K = q_0$ for every $K \geq 1$: the absorbing transcript gives the lower bound, and triggering an arbitrarily long experiment gives the upper bound. Thus the full interval of atomic capacities is realized even with binary branching.

4 One seed for all positive Rényi orders

Fix a rooted controlled history tree with finite nonempty action and output sets at every history and strictly positive transition probabilities. Finite horizons therefore have finitely many histories and policies. Terminal histories may be padded by deterministic steps.

A *causal seed* is a countable probability law on total response strategies: a strategy specifies an output at every history–action pair. It is exact if it realizes the prescribed law under every deterministic adaptive policy at every finite horizon. An *ordinary source* is a countable random variable with a deterministic readout for each such policy and horizon; agreement between the readouts at shared histories is not required. Assume at least one ordinary exact source exists.

All logarithms and entropies are in bits. Write

$$O_\alpha = \inf_{U \text{ ordinary exact}} H_\alpha(U), \qquad C_\alpha = \inf_{U \text{ causal exact}} H_\alpha(U),$$

where $H_\alpha(p) = (1-\alpha)^{-1} \log_2 \sum_i p_i^\alpha$ for $\alpha \neq 1$, H_1 is Shannon entropy, and $H_\infty(p) = -\log_2 \sup_i p_i$. Infinite values are allowed.

Theorem 4.1. *There exists a single exact countable causal seed S such that, simultaneously for every $0 < \alpha < \infty$,*

$$\boxed{O_\alpha \leq C_\alpha \leq H_\alpha(S) \leq O_\alpha + d_\alpha,}$$

where

$$d_\alpha = \frac{\log_2\left(\frac{1}{2} + \frac{2^{-\alpha}}{\alpha}\right)}{1-\alpha} \quad (\alpha \neq 1), \qquad d_1 = \frac{1 + \log_2 e}{2}.$$

Moreover $H_\infty(S) = C_\infty = O_\infty$. Here ∞ denotes the Rényi order, not the time horizon.

4.1 A distributional compactness argument

For the horizon-n transcript law $P_{n,\pi}$, put

$$F_n(t) = \min_\pi P_{n,\pi}\{-\log_2 P_{n,\pi}(X) \leq t\}, \qquad F(t) = \inf_n F_n(t).$$

Zero-probability transcripts carry no mass in this expression. Extending a policy and refining its transcript shows $F_{n+1} \leq F_n$. Each F_n is a CDF; F is nondecreasing and right-continuous. For the latter, use $F(t) = \inf_n \inf_{u>t} F_n(u) = \inf_{u>t} F(u)$.

Let $I_U = -\log_2 p_U(U)$ for any countable source. Deterministic readouts merge atoms, so exactness gives

$$F_{I_U}(t) \leq F_n(t) \quad \text{for all } n, t. \tag{4.1}$$

Consequently $F(t) \to 1$ as $t \to \infty$ whenever an ordinary countable source exists. Thus F is the CDF of a finite nonnegative random variable Z. Let Z_n have CDF F_n; then $Z_n \preceq_{\mathrm{st}} Z$ and $Z_n \Rightarrow Z$.

Finite input. The finite theorem of Section 2 gives a finite exact causal seed at each horizon, with information J_n, for which

$$Z_n \preceq_{\mathrm{st}} J_n \preceq_{\mathrm{st}} Z_n + W, \qquad W = A(1+E), \quad A \sim \mathrm{Bernoulli}(1/2), \quad E \sim \mathrm{Exp}(\ln 2). \tag{4.2}$$

Here A, E, Z_n are independent. The lower comparison is the deterministic-readout inequality.

Lemma 4.2. *There exists an exact countable causal seed S with information J such that*

$$\boxed{Z \preceq_{\mathrm{st}} J \preceq_{\mathrm{st}} Z + W,}$$

with Z and W independent on the right-hand side.

Proof. List the finite greedy atom weights in decreasing order, pad by zeros, and write them as $p_{n,i}$, $i \geq 1$. Extend each finite response strategy arbitrarily to the full infinite tree. As $Z_n \preceq_{\mathrm{st}} Z$, (4.2) gives $J_n \preceq_{\mathrm{st}} Z + W$. For $t \geq 0$, at most $\lfloor 2^t \rfloor$ atoms have weight at least 2^{-t}. Hence

$$\sum_{i > \lfloor 2^t \rfloor} p_{n,i} \leq \mathbb{P}(J_n > t) \leq \mathbb{P}(Z + W > t). \tag{4.3}$$

This is a uniform tail estimate for the *ranked weights*, with no entropy assumption.

The full strategy space is a countable product of finite discrete sets. Diagonal extraction gives a subsequence $n_j \to \infty$ such that, for every i, $p_{n_j,i} \to p_i$ and the associated strategies converge coordinatewise to s_i. Finite partial sums have limits. By (4.3), their tails are uniformly arbitrarily small, so $\sum_i p_i = 1$ and $\sum_i |p_{n_j,i} - p_i| \to 0$.

Give label i weight p_i and response strategy s_i, discarding zero weights. Fix a finite horizon, policy and transcript. The indicator that strategy $s_{n_j,i}$ generates that transcript eventually equals its limit for each fixed i, because it depends on finitely many coordinates. Splitting the weighted sum at a finite rank, and then using the uniform tail estimate, proves exactness of the limit. This works for every finite policy law.

For completeness, $J_{n_j} \Rightarrow J$ follows directly from the same argument. For bounded continuous f, write

$$\mathbb{E} f(J_{n_j}) = \sum_i p_{n_j,i} f(-\log_2 p_{n_j,i}).$$

A fixed positive limiting weight gives convergence of its term; a zero limiting weight gives a term tending to zero in absolute value. Uniformly small rank tails control the remaining sum. Thus the expectations converge to $\sum_{i:p_i>0} p_i f(-\log_2 p_i)$.

Finally $Z_{n_j} \Rightarrow Z$ and $Z_{n_j} + W \Rightarrow Z + W$. First-order stochastic order is closed under weak convergence: test against bounded continuous increasing functions. Passing to the limit in (4.2) proves the lemma. □

4.2 Entropy comparisons

Proof of Theorem 4.1. Take the single seed of Lemma 4.2. For $\alpha \neq 1$, define

$$R_\alpha(Z) = \frac{1}{1-\alpha} \log_2 \mathbb{E} 2^{(1-\alpha)Z}.$$

The stochastic sandwich and independence give

$$R_\alpha(Z) \leq H_\alpha(S) \leq R_\alpha(Z) + \frac{\log_2 \mathbb{E} 2^{(1-\alpha)W}}{1-\alpha}. \tag{4.4}$$

For $0 < \alpha < 1$ these follow by monotonicity of the increasing exponential; infinite moments are permitted. For $\alpha > 1$ the exponential is decreasing, and division by $1 - \alpha < 0$ reverses the inequality once more. Its expectation is strictly positive because Z is finite almost surely.

Since $1-\alpha<1$, the exponential moment of W is finite, and

$$\mathbb{E}2^{(1-\alpha)W}=\frac{1}{2}+\frac{1}{2}\,2^{1-\alpha}\int_0^\infty(\ln 2)\,2^{-\alpha x}\,dx=\frac{1}{2}+\frac{2^{-\alpha}}{\alpha}.$$

For $\alpha=1$, use the nonnegative means directly:

$$\mathbb{E}Z\le H_1(S)\le\mathbb{E}Z+\mathbb{E}W,\qquad \mathbb{E}W=\frac{1}{2}\left(1+\frac{1}{\ln 2}\right)=d_1.$$

This also agrees with the continuous extension of d_α at 1.

For any ordinary exact source U, (4.1) says $Z\preceq_{\mathrm{st}} I_U$. Therefore $R_\alpha(Z)\le H_\alpha(U)$, and $\mathbb{E}Z\le H_1(U)$. Taking the infimum over such U in (4.4) yields $H_\alpha(S)\le O_\alpha+d_\alpha$. The other inequalities follow from inclusion of feasible classes and feasibility of S.

For the endpoint, fix any ordinary exact U. The preceding bound gives $H_\alpha(S)\le H_\alpha(U)+d_\alpha$ for every $\alpha>1$. For a countable probability law, $H_\alpha\to H_\infty$ as $\alpha\to\infty$, whereas $d_\alpha\to 0$. Consequently $H_\infty(S)\le H_\infty(U)$ for every U. Taking the infimum and using $O_\infty\le C_\infty\le H_\infty(S)$ proves equality. No interchange of an infimum with a limit is needed. □

Corollary 4.3 (Simultaneous finiteness)**.** *For the same seed S, and for each $0<\alpha\le 1$,*

$$H_\alpha(S)<\infty\iff \textit{some ordinary exact countable source has finite } H_\alpha.$$

In particular the set of orders with finite Rényi entropy is realized by one causal source, even when its Shannon entropy is infinite.

Proof. The forward implication uses S itself as an ordinary source. The reverse implication follows from the theorem and the finite constant d_α. □

Examples of the bounds. The same S satisfies

$$H_1(S)\le O_1+1.2213475204\ldots,\qquad H_2(S)\le O_2+\log_2(8/5),\qquad H_\infty(S)=O_\infty.$$

These are simultaneous approximation statements, not exact simultaneous minimization at finite orders. The construction does not claim that S also attains the Shannon optimum.

5 Keeping zero-probability branches

The simulation model can retain every branch of a nonnegative normalized kernel. For information and moment expressions, only positive atoms are sampled. Let T^+ be the subtree of histories whose path probability is positive, with zero entries omitted from each local output alphabet. Normalization makes every reduced alphabet nonempty.

Restriction gives a map R from full strategies to reduced strategies: at a positive history, keep a positive output, and replace any zero output by a fixed positive default. Extension gives a map E in the other direction by choosing defaults at all omitted histories. Both maps are measurable, and deterministic when applied to seed labels.

For an exact full law, the event that a positive history is reached and a zero-probability response is selected has probability zero. There are only countably many history-action-output triples. Outside their union, restriction therefore preserves every positive-history cylinder. Induction on the path length shows that R sends every exact full seed to an exact reduced seed. Conversely E sends any exact reduced seed to an exact full seed: every omitted path has a first zero transition, which is never selected along a reached positive history.

Using the same label weights preserves seed entropy in either direction; merging labels after a deterministic map can only decrease a positive-order Rényi entropy. Hence the full and

reduced entropy infima agree, for finite and infinite horizons. The policy laws have the same positive transcript masses, so their information envelopes also agree. Applying the preceding theorems on T^+ and extending the resulting strategies proves their versions on the full tree. Zero branches are retained in the final decoder, although the proof uses the support reduction.

6 Stationary kernels and the square-root regime

Use the centered variables X_i of Section 1; put $a = \max_i \sqrt{\mathbb{E}X_i^2}$, $b = \min_i \sqrt{\mathbb{E}X_i^2}$, and $M = \max_i \|X_i\|_\infty$. The proof applies to arbitrary finitely supported centered increments, even if they are not information variables. The conclusions to prove are

$$C_n = nh + \sqrt{2/\pi}(a-b)\sqrt{n} + o(\sqrt{n}), \tag{6.1}$$

$$C_n = nh + \sqrt{2/\pi}(a-b)\sqrt{n} + O(n^{1/4}\ln n) \quad (b>0). \tag{6.2}$$

6.1 The discrete Bellman envelope

For a policy π, let $S_n^\pi = \sum_{k=1}^n X_{A_k}$, with the fresh increment drawn after the predictable choice A_k. Set

$$G_n(x) = \inf_\pi \mathbb{P}(S_n^\pi \le x), \qquad (Tf)(x) = \min_{j\in\mathcal{A}} \mathbb{E}f(x - X_j).$$

Condition on the first action and outcome. Continuation policies can be chosen separately for each outcome; finite horizon and finite alphabets ensure the minima are attained. Thus

$$G_0(x) = \mathbf{1}_{\{x\ge 0\}}, \qquad G_{n+1} = TG_n. \tag{6.3}$$

The operator is monotone, commutes with spatial translations and addition of constants, and satisfies $\|Tf - Tg\|_\infty \le \|f-g\|_\infty$. Moreover G_n is a CDF: it is the minimum of finitely many CDFs with bounded support. By the chain rule for transcript probabilities,

$$\frac{L_n - nh}{\sqrt{n}} = \int_0^\infty [1 - G_n(x\sqrt{n})]\,dx - \int_0^\infty G_n(-x\sqrt{n})\,dx. \tag{6.4}$$

There is no exchange of an infimum with an integral in this identity. The envelope is formed first, and then its mean is integrated.

6.2 A quantitative comparison for positive least variance

Assume $0 < b \le a$. Write $\phi(z) = (2\pi)^{-1/2}e^{-z^2/2}$ and let Φ be its CDF. Define

$$F_{a,b}(z) = \begin{cases} \dfrac{2b}{a+b}\Phi(z/b), & z \le 0, \\ 1 - \dfrac{2a}{a+b}\Phi(-z/a), & z \ge 0. \end{cases} \qquad u(t,x) = F_{a,b}(x/\sqrt{t}), \quad t>0. \tag{6.5}$$

Both values at zero are $b/(a+b)$. On either half-line, with $s=b$ on the negative side and $s=a$ on the positive side,

$$F'(z) = \frac{2}{a+b}\phi(z/s), \qquad F''(z) = -\frac{2z}{(a+b)s^2}\phi(z/s).$$

Consequently $F \in C^2(\mathbb{R})$, $F''(0) = 0$, and

$$u_t = \min_j \frac{\sigma_j^2}{2} u_{xx}. \tag{6.6}$$

Indeed, $u_{xx} \geq 0$ for $x < 0$, selecting b^2, and $u_{xx} \leq 0$ for $x > 0$, selecting a^2; both sides vanish at zero. No classical third derivative at zero is required.

Fix $M = \max_j \|X_j\|_\infty$ and $\phi_0 = (2\pi)^{-1/2}$. Define

$$A = \frac{M^3\phi_0}{3(a+b)b^2}, \qquad B = \frac{a\phi_0}{2(a+b)}\big(3e^{-1/2} + 3\sqrt{3}\,e^{-3/2}\big),$$
$$\ell = \frac{2\phi_0}{a+b}, \qquad D = \frac{a\phi_0 e^{-1/2}}{a+b}.$$

Lemma 6.1 (One-step defect)**.** *For every $t > 0$,*

$$\|Tu(t,\cdot) - u(t+1,\cdot)\|_\infty \leq At^{-3/2} + \frac{B}{2}t^{-2}. \tag{6.7}$$

Proof. On each open half-line,

$$F'''(z) = \frac{2}{(a+b)s^2}\big((z/s)^2 - 1\big)\phi(z/s).$$

Since $\sup_y |(y^2-1)\phi(y)| = \phi_0$ and F'' is continuous at zero, F'' is globally Lipschitz with constant $2\phi_0/((a+b)b^2)$. Taylor's integral remainder therefore gives, also across zero,

$$\left|\mathbb{E}u(t, x - X_j) - u(t,x) - \frac{\sigma_j^2}{2}u_{xx}(t,x)\right| \leq At^{-3/2}.$$

The linear term vanishes because $\mathbb{E}X_j = 0$. Taking the minimum changes neither this bound nor its uniformity in x. Equation (6.6) identifies the quadratic part with u_t.

For $z = x/\sqrt{t}$,

$$u_{tt}(t,x) = \frac{3zF'(z) + z^2F''(z)}{4t^2}.$$

The elementary maxima $\sup_y |y|\phi(y) = \phi_0 e^{-1/2}$ and $\sup_y |y|^3\phi(y) = 3\sqrt{3}\phi_0 e^{-3/2}$ give $\|u_{tt}(t,\cdot)\|_\infty \leq B/t^2$. Integrating this bound over a unit time step supplies the second remainder $B/(2t^2)$. □

Lemma 6.2 (Explicit CDF comparison)**.** *For $n \geq 1$, $m \geq 1$ and $R \geq 0$,*

$$\sup_z |G_n(z\sqrt{n}) - F_{a,b}(z)| \leq 2\Phi(-R) + \frac{3A}{\sqrt{m}} + \frac{B}{m} + \ell aR\sqrt{\frac{m}{n+m}} + D\ln(1+m/n). \tag{6.8}$$

In particular the left side is $O(n^{-1/4}\sqrt{\log n})$ for $n \geq 2$.

Proof. Let $d = aR\sqrt{m}$ and $\delta = 2\Phi(-R)$. Directly from (6.5),

$$u(m, x-d) - \delta \leq \mathbf{1}_{\{x \geq 0\}} \leq u(m, x+d) + \delta.$$

Apply T^n, preserving order and the translations. By the contraction property and Lemma 6.1, the accumulated defect is at most

$$E_m := \sum_{k=0}^{n-1}\left(\frac{A}{(m+k)^{3/2}} + \frac{B}{2(m+k)^2}\right) \leq \frac{3A}{\sqrt{m}} + \frac{B}{m}.$$

Thus $u(n+m, x-d) - \delta - E_m \leq G_n(x) \leq u(n+m, x+d) + \delta + E_m$. The spatial translation costs at most $\ell d/\sqrt{n+m}$. Also $\sup_x |u_t(t,x)| \leq D/t$, so changing time from $n+m$ to n costs at most $D\ln(1+m/n)$. This proves (6.8). Choose $m = \sqrt{n}$, $R = \sqrt{\ln n}$, and use $2\Phi(-R) \leq e^{-R^2/2}$ to obtain the stated order. □

6.3 The mean: uniform tails, an explicit error bound, and the coefficient

Under every policy, the increments are martingale differences in $[-M, M]$. Conditional Hoeffding's bound and iteration give $\mathbb{E}e^{\theta S_n^\pi} \le e^{n\theta^2 M^2/2}$. Markov's inequality, optimized at $\theta = x/(M^2\sqrt{n})$ and applied also to $-S_n^\pi$, yields, for $x > 0$,

$$1 - G_n(x\sqrt{n}) \le e^{-x^2/(2M^2)}, \qquad G_n(-x\sqrt{n}) \le e^{-x^2/(2M^2)}. \tag{6.9}$$

These bounds are uniform in both n and the adaptive policy. The same bounds hold for $1 - F_{a,b}(x)$ and $F_{a,b}(-x)$ since $a \le M$. Dominated convergence in (6.4) is now justified. Moreover, if e_n denotes the right side of (6.8), then, for every $r > 0$,

$$\left|\frac{L_n - nh}{\sqrt{n}} - \mu_{a,b}\right| \le 2re_n + \frac{4M^2}{r}e^{-r^2/(2M^2)}, \tag{6.10}$$

where $\mu_{a,b}$ is the mean of $F_{a,b}$. Split the CDF-distance integral at $\pm r$ to obtain this inequality, using $\int_r^\infty e^{-x^2/(2M^2)}dx \le (M^2/r)e^{-r^2/(2M^2)}$. With $r = M\sqrt{2\ln n}$, Lemma 6.2 makes (6.10) $O(n^{-1/4}\ln n)$.

Finally, $\int_0^\infty \Phi(-x/s)\,dx = s/\sqrt{2\pi}$ gives

$$\mu_{a,b} = \frac{2a^2 - 2b^2}{(a+b)\sqrt{2\pi}} = \sqrt{\frac{2}{\pi}}(a - b). \tag{6.11}$$

Equations (6.10), (6.11) and (1.1) prove both claims of Theorem 1.3(1) when $b > 0$. In particular, (6.8) and (6.10), plus $c_\star/\sqrt{n}$, are an explicit finite-n bound for the normalized optimal entropy error.

6.4 Zero least variance: a controlled perturbation argument

Assume $b = 0 < a$. For $\varepsilon > 0$, replace every increment by $X_j + \varepsilon\xi$, where each fresh ξ is an independent symmetric sign. The controller is allowed to observe the pair of outcomes. The Bellman state is still the partial sum: the conditional future laws depend only on the next action, so observing the pair rather than its sum changes no value. Let G_n^ε be this envelope. Its extreme standard deviations are $a_\varepsilon = \sqrt{a^2 + \varepsilon^2}$ and $b_\varepsilon = \varepsilon$; the previous proof applies to this bounded, centered family.

For every $\eta > 0$ and $x \in \mathbb{R}$,

$$G_n^\varepsilon((x - \eta)\sqrt{n}) - \varepsilon^2/\eta^2 \le G_n(x\sqrt{n}) \le G_n^\varepsilon((x + \eta)\sqrt{n}) + \varepsilon^2/\eta^2. \tag{6.12}$$

To verify this, couple the original and perturbed sums using the same actions and original outcomes. Their difference is $\varepsilon\sum_{k=1}^n \xi_k$, whose tail beyond $\eta\sqrt{n}$ is at most ε^2/η^2. An original policy can be used in the perturbed system by ignoring the signs. Conversely, a perturbed policy can be run in the original system using independent auxiliary signs. Such randomization does not improve an infimum of event probabilities: conditional on its entire independent random tape it is a deterministic causal policy, so its event probability is an average of probabilities of admissible policies. These two comparisons, with shifted thresholds, prove (6.12).

First let $n \to \infty$ for fixed ε, η, then set $\eta = \sqrt{\varepsilon}$ and let $\varepsilon \downarrow 0$. The explicit profiles tend to the continuous CDF

$$F_{a,0}(x) = \begin{cases} 0, & x \le 0, \\ 2\Phi(x/a) - 1, & x > 0. \end{cases}$$

Equation (6.12) proves $G_n(x\sqrt{n}) \to F_{a,0}(x)$ at every x. The original-family bound (6.9) still holds, so dominated convergence gives mean $a\sqrt{2/\pi}$. This proves (6.1) for $b = 0 < a$, without asserting the remainder (6.2) in that case. If $a = b = 0$, every increment is zero almost surely, $G_n = G_0$, and (1.1) immediately gives (6.1). This finishes the proof.

6.5 Universality of near-optimal seed information

Corollary 6.3 (Universality of asymptotically optimal seeds)**.** *Suppose $a > 0$. Let $J_n = -\log_2 w_n(S_n)$ be the self-information of an exact countable seed with finite entropy, and suppose $\mathbb{E}J_n - C_n = o(\sqrt{n})$. Then the law of $(J_n - nh)/\sqrt{n}$ converges in W_1 to the law with CDF $F_{a,b}$ (with the half-normal convention when $b = 0$).*

Proof. Every seed atom is no heavier than any transcript to which it is mapped. Thus its self-information stochastically dominates each policy transcript, and its CDF is at most their lower envelope. If Z_n has that envelope CDF, then $Z_n \preceq_{\mathrm{st}} J_n$, and $W_1(\mathcal{L}(J_n), \mathcal{L}(Z_n)) = \mathbb{E}J_n - L_n$. By (1.1) this is $o(\sqrt{n})$. The CDF convergence and uniform tail bound above give W_1 convergence of $(Z_n - nh)/\sqrt{n}$ to $F_{a,b}$. The triangle inequality proves the claim. □

The profile is not the limiting transcript law of one policy when $a > b$: every centered transcript has mean zero and the uniform tail bound forces any weak limit to have mean zero. The envelope profile has strictly positive mean. Different thresholds may need different minimizing policies. Treating the whole envelope as one policy law would invalidate the argument.

7 The logarithmic regime with bounded remainder

Let $X_1, \dots, X_d$ have finitely supported laws satisfying

$$\mathbb{E}X_i = 0, \qquad \mathbb{E}X_i^2 = v > 0.$$

Set $\kappa_i = \mathbb{E}X_i^3$, $\kappa_- = \min_i \kappa_i$, $\kappa_+ = \max_i \kappa_i$, and

$$A_i f(x) = \mathbb{E}f(x - X_i), \qquad Tf = \min_i A_i f, \qquad Uf = \max_i A_i f.$$

Both operators preserve order and commute with translations. The operator U is positively homogeneous and subadditive, and

$$|Tf - Tg| \le U|f - g|, \qquad |Uf - Ug| \le U|f - g|. \tag{7.1}$$

Let $H = \mathbf{1}_{[0,\infty)}$, $G_n = T^n H$, and let ν_n be the probability law with CDF G_n. The mean of this envelope law is $e_n = \int_{\mathbb{R}}(H - G_n)$. The family is fixed throughout; constants are permitted to depend on it. Write ϕ, Φ for the standard Gaussian density and CDF.

Theorem 7.1. *With $c = (\kappa_+ - \kappa_-)/(3v\sqrt{2\pi e})$, there is a finite constant K such that for all integers $n \ge 1$,*

$$\boxed{W_1(\nu_n, N(c\ln n, vn)) \le K, \qquad |e_n - c\ln n| \le K.} \tag{7.2}$$

No lattice or aperiodicity assumption is needed.

Corollary 7.2. *For any finite stationary kernel with common action entropy h and common positive information variance v, in bits,*

$$\boxed{C_n = nh + \frac{\kappa_+ - \kappa_-}{3v\sqrt{2\pi e}}\ln n + O(1),}$$

where $\kappa_i = \mathbb{E}(-\log_2 P_i(Y_i) - h)^3$. The same expansion holds for the envelope mean and for exact seeds of entropy within a bounded additive amount of that mean.

Proof of the corollary from Theorem 7.1. The centered transcript-information envelope is G_n, and $L_n = nh + e_n$. Use the finite-tree sandwich (1.1). □

7.1 Uniform Gaussian amplification

For $t > 0$ denote the $N(0, vt)$ density by

$$\gamma_t(x) = \frac{1}{\sqrt{2\pi vt}} e^{-x^2/(2vt)}.$$

Let P be convolution with $N(0, v)$, so $P\gamma_t = \gamma_{t+1}$. This is a comparison operator, not an additional available action.

Lemma 7.3 (One-step Gaussian defect)**.** *There is $0 \le D < \infty$ such that for all $t \ge 1$ and $x \in \mathbb{R}$,*

$$|U\gamma_t(x) - \gamma_{t+1}(x)| \le Dt^{-3/2}\gamma_{8t}(x). \tag{7.3}$$

Proof. Let $M = \max_i \max\{|y| : y \in \operatorname{supp} X_i\}$. Centered Taylor expansion and equality of the second moments give

$$\left| A_i\gamma_t - \gamma_t - \frac{v}{2}\gamma_t'' \right|(x) \le \frac{\mathbb{E}|X_i|^3}{6} \sup_{|a|\le M} |\gamma_t'''(x-a)|.$$

Also $\partial_t\gamma_t = (v/2)\gamma_t''$, whence

$$\left| \gamma_{t+1} - \gamma_t - \frac{v}{2}\gamma_t'' \right|(x) \le \frac{1}{2} \sup_{t\le s\le t+1} |\partial_s^2\gamma_s(x)|.$$

For finite constants D_3, D_4, uniformly in $t \ge 1$,

$$\sup_{|a|\le M} |\gamma_t'''(x-a)| \le D_3 t^{-3/2}\gamma_{8t}(x), \tag{7.4}$$

$$\sup_{t\le s\le t+1} |\partial_s^2\gamma_s(x)| \le D_4 t^{-2}\gamma_{8t}(x). \tag{7.5}$$

Here is a direct justification of uniformity in x. With $z = x/\sqrt{vt}$, $b = a/\sqrt{vt}$, one has $|b| \le M/\sqrt{v}$ and $(z-b)^2 \ge z^2/2 - b^2$. Gaussian spatial derivatives are a polynomial times their Gaussian density. Thus the ratio in (7.4), after removing $t^{-3/2}$, is bounded by a constant times $(1+|z|)^3 e^{-3z^2/16}$. For (7.5), use $\partial_s^2\gamma_s = (v^2/4)\gamma_s''''$ and $1 \le s/t \le 2$; the analogous ratio is bounded by a constant times $(1+|z|)^4 e^{-3z^2/16}$. Both suprema are finite. Finally $t^{-2} \le t^{-3/2}$ and $|\max_i a_i - b| \le \max_i |a_i - b|$, proving (7.3). □

Lemma 7.4 (Bounded integrated amplification)**.** *Choose $A \ge 1$ with $3D/\sqrt{A} \le 1/2$, where D is from Lemma 7.3. Then for every $t \ge A$, every $m \in \mathbb{R}$ and every integer $n \ge 0$,*

$$\boxed{\int_{\mathbb{R}} U^n[\gamma_t(\cdot - m)](x)\,dx \le 2.} \tag{7.6}$$

For example $A = \max\{1, 36D^2\}$ is admissible.

Proof. Translation invariance reduces the proof to $m = 0$. Positivity, subadditivity, (7.1), and $P^r\gamma_t = \gamma_{t+r}$ give the pointwise telescoping bound

$$U^n\gamma_t \le \gamma_{t+n} + D\sum_{r=0}^{n-1} (t+r)^{-3/2} U^{n-1-r}\gamma_{8(t+r)}. \tag{7.7}$$

For clarity, obtain this by comparing $U^n\gamma_t$ with $U^{n-1}\gamma_{t+1}$, then $U^{n-2}\gamma_{t+2}$, and so forth; each difference is bounded by the appropriate iterate of U applied to $|U\gamma_{t+r} - \gamma_{t+r+1}|$.

Use strong induction on n, simultaneously for every real $t \geq A$. The assertion at $n = 0$ is the Gaussian integral, equal to one. In (7.7), every remaining horizon is less than n and every Gaussian time $8(t + r)$ is at least A. The induction hypothesis therefore bounds its integral by two. Since

$$\sum_{r=0}^{\infty} (t + r)^{-3/2} \leq t^{-3/2} + \int_0^{\infty} (t + s)^{-3/2}\, ds \leq \frac{3}{\sqrt{t}},$$

the integrated right side of (7.7) is at most $1 + 6D/\sqrt{t} \leq 2$. This closes the induction. There is no assumption that the supremum over all horizons is already finite. □

A general stability consequence. If $|f - g| \leq b\gamma_t(\cdot - m)$ with $b \geq 0$ and $t \geq A$, then

$$\|T^n f - T^n g\|_1 \leq 2b \qquad (n \geq 0). \tag{7.8}$$

This follows from (7.1) and Lemma 7.4. The same holds for sums of such majorants, with $2\sum b$ on the right. A further consequence, for $f \geq 0$ integrable, is $\int U^n(\gamma_t * f) \leq 2\,\|f\|_1$ when $t \geq A$: take the supremum over policies inside the nonnegative integral against f, and apply Tonelli and (7.6).

7.2 The resonant corrector

Define

$$f(z) = \min_i \left[-\frac{\kappa_i}{6v^{3/2}} (z^2 - 1)\phi(z) \right], \tag{7.9}$$

$$a_0 = \int_{\mathbb{R}} f(z)\, dz = -\frac{\kappa_+ - \kappa_-}{3v^{3/2}} \phi(1) = -\frac{c}{\sqrt{v}}, \tag{7.10}$$

$$g(z) = 2(a_0\phi(z) - f(z)), \qquad J(z) = \int_{-\infty}^{z} g(y)\, dy, \tag{7.11}$$

$$B(z) = \phi(z) \int_0^z \frac{J(y)}{\phi(y)}\, dy. \tag{7.12}$$

Indeed $\int (z^2 - 1)\phi = 0$ and $\int |z^2 - 1|\phi = 4\phi(1)$, proving (7.10). Since $\int g = 0$, formula (7.12) decays at both ends and solves

$$B'' + zB' + B = 2(a_0\phi - f). \tag{7.13}$$

Regularity and tails, explicitly. The function f is continuous, smooth off $\{-1, 1\}$, and has only derivative jumps there. Thus $B \in C^2$, with locally absolutely continuous second derivative. For $z < -1$ and $z > 1$, respectively,

$$\frac{J(z)}{\phi(z)} = 2a_0 \frac{\Phi(z)}{\phi(z)} - \frac{\kappa_+ z}{3v^{3/2}}, \qquad \frac{J(z)}{\phi(z)} = -2a_0 \frac{1 - \Phi(z)}{\phi(z)} - \frac{\kappa_+ z}{3v^{3/2}}.$$

The elementary Gaussian tail inequality $1 - \Phi(z) \leq \phi(z)/z$ for $z > 0$, and its reflected version, show that the integral in (7.12) is $O(1 + z^2)$. The equation $B' + zB = J$ and (7.13), followed by its almost-everywhere derivative, then show that B, B', B'', B''' are bounded by a polynomial times ϕ on each tail. All their polynomially weighted absolute integrals are finite. In particular $B \in W^{3,1}(\mathbb{R})$. No fourth classical derivative of B is required.

For $t \geq 1$, put

$$z = z_t(x) = \frac{x - c \ln t}{\sqrt{vt}}, \qquad p_t(x) = \Phi(z), \quad q_t(x) = t^{-1/2} B(z), \quad u_t = p_t + q_t. \tag{7.14}$$

The function u_t need not be a CDF; it is used only as a bounded comparison profile. Its limiting values are zero and one. Direct differentiation using (7.13) gives

$$\partial_t u_t - \frac{v}{2}\partial_x^2 u_t = t^{-3/2} f(z) - \frac{c}{\sqrt{v}}\, t^{-2} B'(z). \tag{7.15}$$

The translation $c \ln t$ absorbs the resonant Gaussian mode exactly.

7.3 Pointwise discrete residual

The following estimate closes the analytic input needed for bounded propagation. All constants below depend only on the fixed increment family. Put

$$E_0(z) = (1+z^2)^4 e^{-z^2/8}.$$

Lemma 7.5. *There is $K_0 < \infty$ such that for every real $t \geq 1$ and $x \in \mathbb{R}$,*

$$|u_{t+1}(x) - T u_t(x)| \leq K_0 t^{-2} E_0(z_t(x)). \tag{7.16}$$

Proof. Write $p_t(x) = \Phi(z_t(x))$ and $q_t(x) = t^{-1/2} B(z_t(x))$. We first specify the uniform Gaussian bounds used in Taylor's formula. The corrector construction gives

$$|B^{(j)}(z)| \leq C(1+|z|)^{j+2}\phi(z), \qquad 0 \leq j \leq 3,$$

where the last derivative is interpreted almost everywhere. For $j = 0$ this follows from the integral representation and the Gaussian tail bound; $B' = J - zB$, $B'' = 2(a_0\phi - f) - zB' - B$, and the almost-everywhere derivative of the last identity give $j = 1, 2, 3$. On the compact interval $[-1, 1]$ all these bounds follow by boundedness, enlarging C if necessary.

Let M bound every increment. For a spatial shift $|y| \leq M$, the standardized argument changes from $z = z_t(x)$ by at most $M/\sqrt{v}$. For $t \leq s \leq t+1$ the time-changed argument is

$$z_s(x) = \sqrt{t/s}\, z + \frac{c \ln(t/s)}{\sqrt{vs}}.$$

Here $\sqrt{t/s} \in [1/\sqrt{2}, 1]$, and the additive term is uniformly bounded. The inequality $(rz+d)^2 \geq r^2 z^2/2 - d^2$, valid for all real z, d, shows that any polynomial of degree at most eight times $\phi(rz+d)$ is bounded by a constant times $E_0(z)$ in these windows. For spatial shifts use $r = 1$; for time changes use $r \geq 1/\sqrt{2}$. Thus the constants in the following derivative estimates are uniform in t, x and throughout the indicated Taylor segments:

$$\begin{aligned}
\sup_{|y|\leq M} |p_t^{(4)}(x-y)| &\leq C t^{-2} E_0(z), \\
\operatorname*{ess\,sup}_{|y|\leq M} |q_t^{(3)}(x-y)| &\leq C t^{-2} E_0(z), \\
\sup_{s\in[t,t+1]} |\partial_s^2 u_s(x)| &\leq C t^{-2} E_0(z).
\end{aligned}$$

For the last line, differentiate $z_t' = -z_t/(2t) - c/(\sqrt{v}\, t^{3/2})$ once more. The resulting terms involve only ϕ, ϕ', B, B', B'', multiplied by polynomials of degree at most two and by powers no larger than t^{-2} for $t \geq 1$. The preceding window estimate applies to every term.

Centered spatial Taylor expansion, first to order three and then to order two, gives

$$\begin{aligned}
A_i p_t &= p_t + \frac{v}{2} p_t'' - \frac{\kappa_i}{6} p_t''' + r_{i,t}, \\
A_i q_t &= q_t + \frac{v}{2} q_t'' + s_{i,t},
\end{aligned}$$

with $|r_{i,t}(x)| + |s_{i,t}(x)| \le Ct^{-2}E_0(z)$. Indeed the two integral remainders are bounded by $\mathbb{E}|X_i|^4/24$ and $\mathbb{E}|X_i|^3/6$ times the respective derivative suprema. The C^2 regularity and local absolute continuity of B'' justify Taylor's formula across the switching points ± 1. Taking the minimum over i is Lipschitz in the maximum norm, so

$$Tu_t = u_t + \frac{v}{2}u_t'' + t^{-3/2}f(z) + O(t^{-2}E_0(z)).$$

On the other hand, the corrector equation gives exactly

$$\partial_t u_t - \frac{v}{2}u_t'' = t^{-3/2}f(z) - \frac{c}{\sqrt{v}}t^{-2}B'(z).$$

Time Taylor expansion on $[t, t+1]$ and the time-derivative estimate above yield

$$u_{t+1} = u_t + \frac{v}{2}u_t'' + t^{-3/2}f(z) + O(t^{-2}E_0(z)).$$

Subtracting proves the claim. □

Let $D_E = 64^4 e^{-63/16}$. Maximizing $(1+z^2)^4 e^{-z^2/16}$ shows that it is at most D_E. Consequently, with $b_0 = K_0 D_E \sqrt{16\pi v}$,

$$|u_{t+1} - Tu_t|(x) \le b_0 t^{-3/2}\gamma_{8t}(x - c\ln t). \tag{7.17}$$

The mass of this majorant is summable over $t = T, T+1, \ldots$. The amplification lemma ensures that summability survives every remaining Bellman iteration.

7.4 Completion of the bounded-remainder proof

Fix any real $T \ge \max\{A, 2\}$ once and for all, independently of n. The initial error has a Gaussian majorant

$$|H(x) - u_T(x)| \le b_1\gamma_{s_0}(x - c\ln T) \tag{7.18}$$

for some finite b_1 and some $s_0 \ge A$. To verify this, choose $s_0 = 8T$. On each sufficiently distant tail, $H - \Phi(z_T)$ is a Gaussian tail and $B(z_T)$ is bounded by a polynomial times $\phi(z_T)$. Dividing by the broader Gaussian $\gamma_{8T}(x - c\ln T)$ gives a bounded ratio tending to zero. On the remaining compact interval the numerator is bounded and the denominator has a positive minimum. The jump of H at zero does not affect this pointwise bound.

Using (7.1) repeatedly, followed by (7.6), (7.17), and (7.18), gives

$$\begin{aligned}
\|G_n - u_{T+n}\|_1 &\le \int U^n|H - u_T| + \sum_{j=0}^{n-1}\int U^{n-1-j}|u_{T+j+1} - Tu_{T+j}| \\
&\le 2b_1 + 2b_0\sum_{j=0}^{n-1}(T+j)^{-3/2} \\
&\le 2b_1 + \frac{6b_0}{\sqrt{T}} =: K_1.
\end{aligned} \tag{7.19}$$

This is uniform in n. Moreover

$$\|u_t - \Phi(z_t)\|_1 = \sqrt{v}\,\|B\|_1.$$

Finally the Gaussians with means $c\ln(n+T)$ and $c\ln n$, and variances $v(n+T)$ and vn, can be coupled using the same standard normal variable. Their W_1 distance is at most

$$|c|\ln(1 + T/n) + \sqrt{2v/\pi}\,(\sqrt{n+T} - \sqrt{n}),$$

which tends to zero and is bounded for $n \ge 1$. The one-dimensional formula $W_1(\mu,\nu) = \int |F_\mu - F_\nu|$ and (7.19) prove the first part of (7.2). The difference of means is bounded by W_1, proving the second. This proves Theorem 7.1. □

7.5 An explicit information pair

Take

$$P = (1/4, \underbrace{1/16, \ldots, 1/16}_{12}), \qquad Q = (\underbrace{1/8, \ldots, 1/8}_{6}, \underbrace{1/32, \ldots, 1/32}_{8}).$$

For P, centered information is $-3/2$ with probability $1/4$ and $1/2$ with probability $3/4$. For Q, it is $-1/2$ with probability $3/4$ and $3/2$ with probability $1/4$. Direct calculation yields

$$h = 7/2, \qquad v = 3/4, \qquad \kappa_P = -3/4, \quad \kappa_Q = 3/4.$$

Therefore

$$\boxed{C_n = \frac{7}{2}n + \frac{2}{3\sqrt{2\pi e}} \ln n + O(1).}$$

This is the optimum over all exact causal seeds, not the cost of a restricted block architecture. The coefficient is $0.161313816346\ldots$ bits per unit of $\ln n$. The example's dyadic structure is not used in proving this asymptotic. A separate dyadic refinement theorem in the repository additionally identifies $C_n = L_n$ at every horizon.

7.6 What bounded amplification resolves

A summable local residual alone would be insufficient if its propagation could grow with the remaining horizon. Lemma 7.4 controls precisely that growth on the Gaussian majorants produced by the corrector. A fixed starting time then suffices. No smoothness of the actual envelope G_n, nonlattice hypothesis, or prior logarithmic bound on its width is needed in this proof. In particular, when the third moments all agree, $c = 0$ and the same argument proves bounded excess directly.

8 Formalization, reproducibility, and scope

The formal development is available at

https://github.com/jendamikulik/finite-Shannon-theorem.

The revision used for this paper is

`5741ee193bd5ff7f4addd4c09b2d65edfe238c20`.

It pins Lean `4.35.0-rc3` and Mathlib revision `7d6757bc18680044fc0ff6efc8ee20494b408556`. The import root is `CausalSeed/Main.lean`. The repository supplies the commands

```
lake build
lake env lean scripts/AxiomAudit.lean
```

for compilation and enumeration of theorem dependencies. The recorded `AXIOMS.txt` lists 2528 project theorems; all recorded dependencies are among `propext`, `Classical.choice`, and `Quot.sound`, and seven theorems use none. It reports no nonstandard axioms and no project axiom declarations. The audit includes `Kernel.cost_bounded` and `LogGauss.bounded_L1`. These are the recorded results for the cited revision, not a report of a new build performed while typesetting this manuscript.

8.1 Statement-to-source map

All names below lie under `CausalSpectrum`. Paths are relative to `CausalSeed/`. Names are provided instead of unstable line numbers.

Mathematical statement	Module and principal declaration
Finite Shannon bound	`Entropy.lean`: `one_seed_shannon`, `causal_realization`
Information converse	`Penalty.lean`: `entropy_ge_envelope`; `Entropy.lean`: `seedCDF_le_Fstar`
CDF/transport identity	`GapW1.lean`: `wasserstein_gap_W1`; `Wasserstein.lean`: `OT.W1_eq_cdfDist`
Projective identity	`EntropyLimit.lean`: `Completion.projective_entropy_identity`, `Completion.entropy_attained`
Infinite sandwich	`InfiniteEnvelope.lean`: `Completion.infinite_envelope_sandwich`
Atomic capacity	`AtomicCapacity.lean`: `Completion.causal_atomic_capacity_attained`, `Completion.countable_seed_iff_atomQ_one`
Capacity from the envelope	`AtomicSpectrum.lean`: `atomQ_eq_lim_FInf`
Simultaneous Rényi bound	`RenyiLimitMeasure.lean`: `RenyiInf.one_seed_all_orders_meas`
Zero branches	`ZeroClosure.lean`; `ZeroBranchTransfer.lean`; `ZeroBranchOrdinary.lean`
Square-root asymptotic	`SqrtGeneral.lean`: `Kernel.cost_tendsto`; `SqrtMain.lean`: `Kernel.cost_rate`
Universality	`SqrtW1.lean`: `Kernel.universality_W1`
Corrector and residual	`LogCorrector.lean`: `LogGauss.cB_ode`; `LogResidual.lean`: `LogGauss.residual_pointwise`
Gaussian amplification	`LogBounded.lean`: `LogGauss.amp_bounded`
Bounded logarithmic remainder	`LogBounded.lean`: `LogGauss.bounded_L1`, `LogGauss.bounded_mean`, `Kernel.cost_bounded`
Explicit dyadic pair	`LogBounded.lean`: `Kernel.LogPair.cost_bounded`

8.2 Definitions and the verification boundary

The finite objects `Tree`, `Policy`, `Strategy`, `run`, and `leafMass` are defined in `Spectrum.lean`. The functions `Fstar` and `envelopeMean` in `Entropy.lean` represent the information envelope and its mean. The infinite model uses `ITree`, `Completion.CountableSeed`, and its predicates `ExactTo` and `Exact`. Ranking and finite restriction are connected to arbitrary countable seed alphabets by the bridge modules. The infinite implementation uses extended entropy in nats; `Kernel.costBits` and the envelope bridges convert to bits.

An axiom audit checks logical dependencies of the formal statements. Reading the definitions and their hypotheses is still necessary to assess whether those statements express the intended simulation problem. In particular, the minimum over transcript CDFs is taken pointwise; it is not assumed to be the law of one policy. The fixed-kernel constants in Section 7 are existential and need not be uniform as the variance or the kernel varies. Formal verification does not establish historical priority or numerical practicality.

8.3 Open questions

The bounded logarithmic remainder need not be treated as a convergent constant. This paper proves neither convergence of $C_n - nh - c \ln n$ nor an expansion for its successive differences.

Even if the envelope mean has a limiting constant, the bounded entropy gap must still be controlled to transfer such a limit to C_n.

Other questions in the accompanying development include the optimal universal additive constants and the exact supremum G_* of the causal minus ordinary Shannon optimum. Its recorded bounds are $2/9 \le G_* \le c_\star$; the endpoint behavior of exact penalties at G_* is separate from the asymptotic theorem here. For unequal action entropies, an $O(\ln n)$ comparison with the maximal-entropy subfamily does not transfer a logarithmic coefficient. We therefore impose common entropy throughout the sharp logarithmic theorem.

Acknowledgment of assistance. AI systems, including ChatGPT and Grok, were used during the development of mathematical arguments, code, formalization, and exposition. The cited source revision and the explicit statements above delimit the mathematical claims; the use of these systems is not evidence of their correctness.

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